Design and CFD Analysis of Radiator by Using Ansys

PRESENTED BY ;

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INTRODUCTION

- Radiators are essential for cooling engines, preventing overheating and ensuring performance and durability.
- Modern radiator design balances heat transfer and pressure drop, critical for advanced engines, including electric and hybrid models.
- CFD tools like ANSYS enable precise simulations, optimizing radiator designs before physical prototyping.
- This study uses ANSYS to examine variables (material, fin density, tube arrangement) affecting radiator efficiency.
- Results aim to support sustainable automotive design, aligning with fuel efficiency and emission reduction goals.

OBJECTIVE

The objective of this study is to conduct a comprehensive analysis of an automobile radiator using Computational Fluid Dynamics (CFD) in ANSYS to optimize its thermal and fluid performance. The study aims to evaluate the radiator's heat transfer efficiency and temperature distribution under various operational conditions while analyzing fluid flow patterns to identify turbulence and areas for improvement. Key design parameters, such as fin density, tube dimensions, and material selection, will be refined to enhance cooling efficiency and minimize pressure drop. Additionally, the structural integrity of the radiator will be assessed to ensure it can withstand thermal and mechanical stresses. Ultimately, this research seeks to validate radiator performance under real-world conditions, contributing to the development of efficient, sustainable radiator designs that align with modern automotive requirements.

PROBLEM DEFINITION

Problem Definition : Automobile radiators must efficiently dissipate engine heat to prevent overheating, posing challenges in balancing heat transfer, pressure drop, and weight, especially with modern engine complexities. Traditional design methods are costly and time-consuming, necessitating an effective virtual approach using CFD tools to optimize radiator performance while meeting sustainability goals.

Scope of the Project : The project involves developing a detailed CFD model of an automobile radiator in ANSYS to analyze heat transfer and fluid dynamics. Key design parameters will be optimized for performance, structural integrity will be assessed, and real-world performance will be validated under varying conditions to ensure reliability in modern automotive applications.

INFORMATION GATHERING

To effectively optimize an automobile radiator using CFD, this project will gather extensive information across several key areas. First, data on thermal properties of materials, such as thermal conductivity and heat capacity, are essential for accurate heat transfer modeling. Similarly, fluid properties of the coolant, including density, viscosity, and specific heat capacity, are needed to analyze fluid dynamics and pressure drop. We will also collect operational data, such as engine temperature ranges, coolant flow rates, and environmental conditions (like ambient temperature and vehicle speed), to simulate real-world conditions effectively.

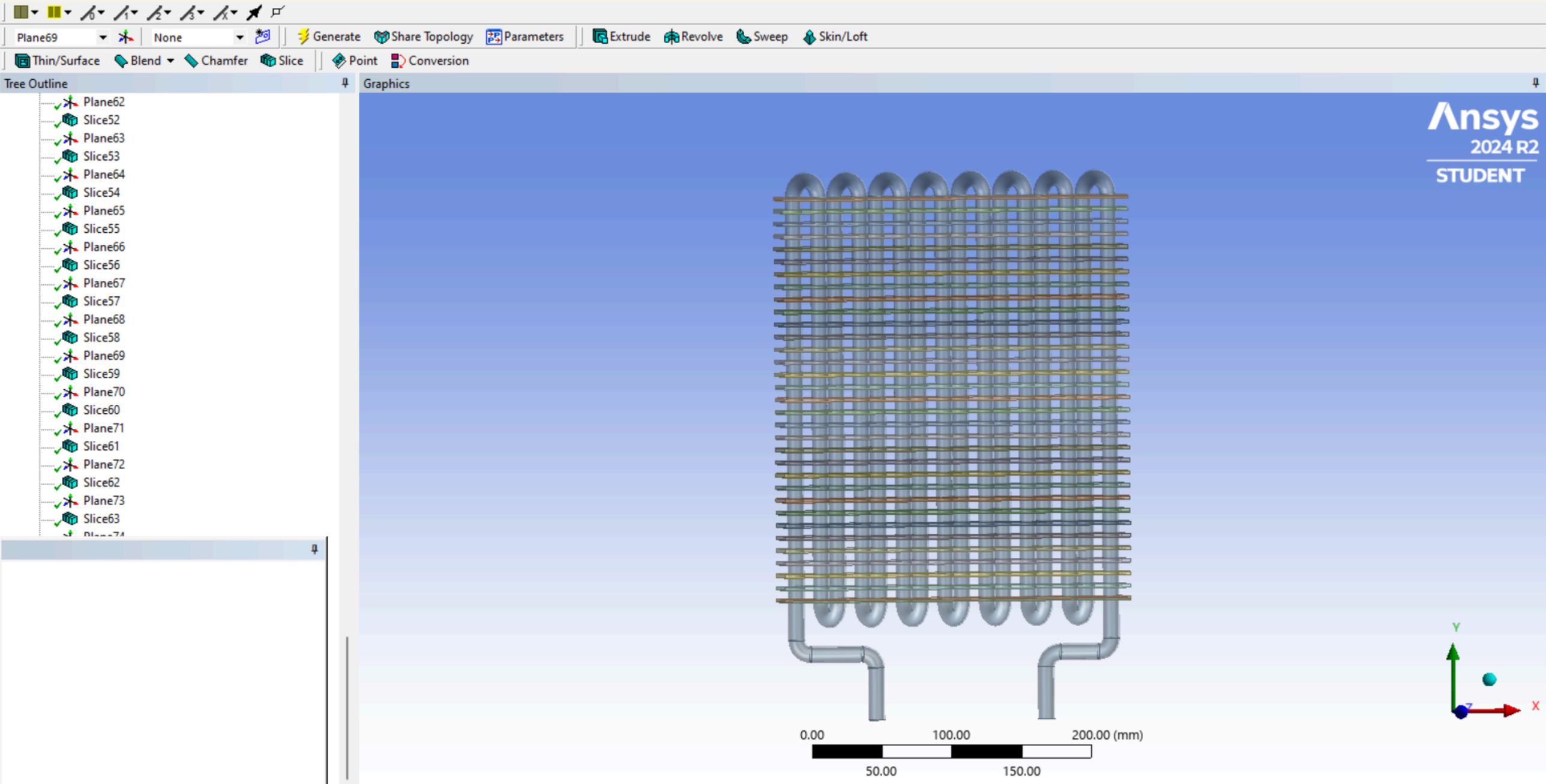
To guide design optimization, benchmarking data on industry-standard radiator designs—covering aspects like fin density, tube dimensions, and material choices—will be reviewed to understand current practices and opportunities for improvement. Industry standards related to fuel efficiency and emissions will inform our objectives, ensuring alignment with modern sustainability goals. Lastly, recent literature on CFD advancements and radiator technologies will be studied to select effective simulation techniques, configurations, and performance metrics, helping to create an accurate and optimized design.

MODELLING AND DESIGN

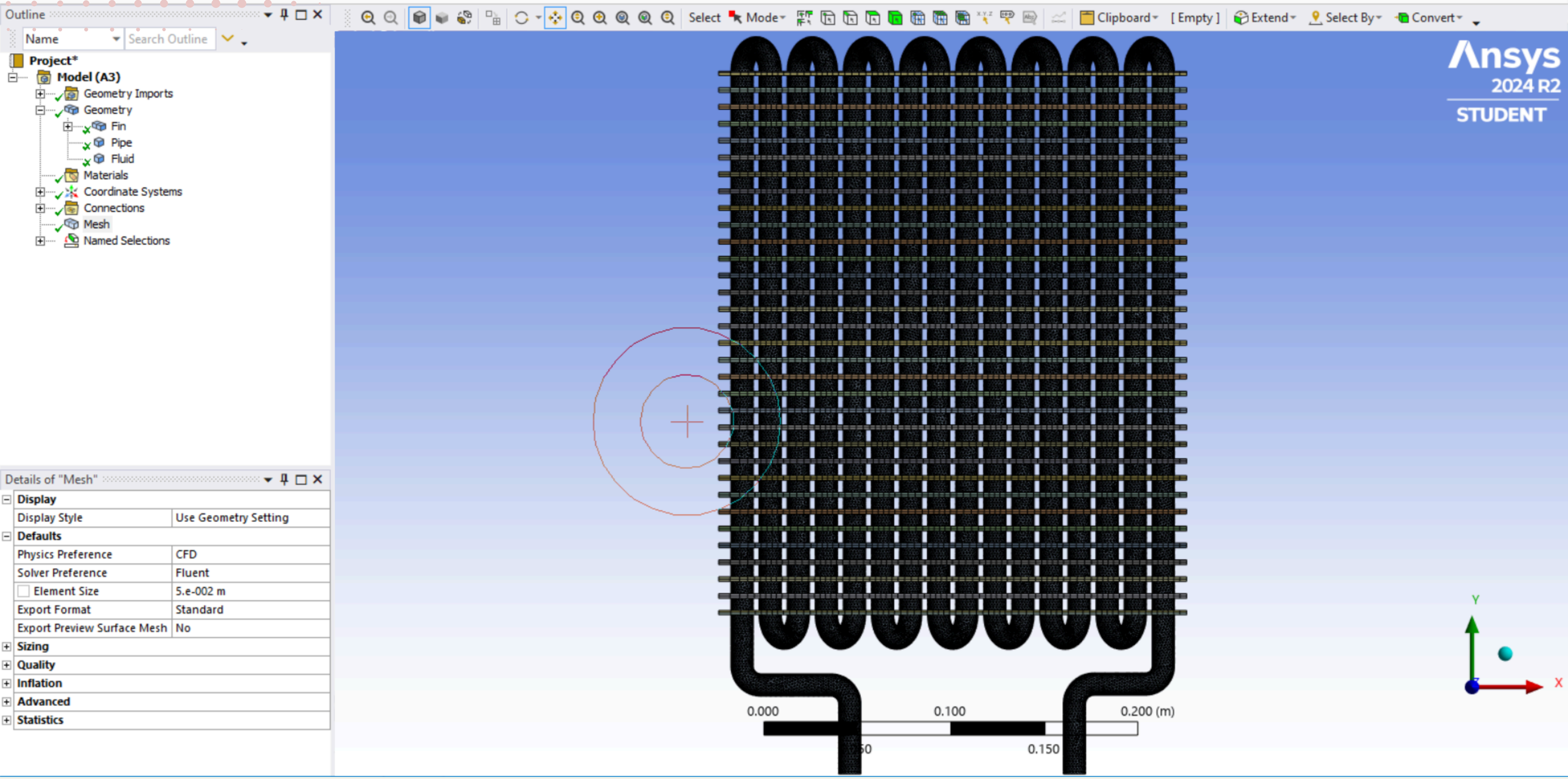
The modeling and design phase will involve creating a detailed 3D model of the radiator in ANSYS, capturing critical features such as fins, tubes, and headers. Key design parameters, including fin density, tube dimensions, and material selection, will be incorporated to represent realistic radiator geometries accurately. The model will be refined to ensure an optimal balance between complexity and computational efficiency, focusing on areas that significantly affect thermal and fluid performance.

To simulate real-world operating conditions, boundary conditions will be defined based on engine temperature ranges, coolant flow rates, and external airflow. Material properties will also be carefully selected to represent thermal conductivity and strength accurately. The model will then undergo meshing, with a high-resolution mesh applied to areas of interest (e.g., fins and tubes) to capture detailed flow and temperature gradients. This precise modeling approach will enable accurate CFD simulations to evaluate and enhance the radiator's thermal efficiency and structural stability.

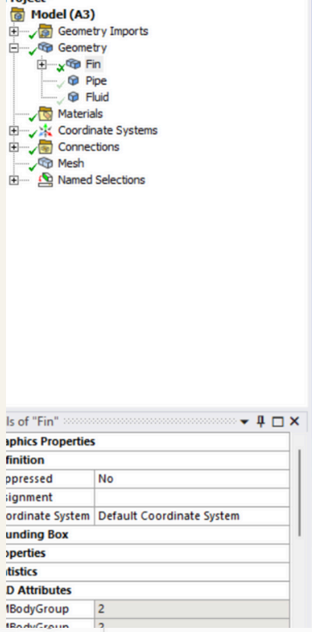
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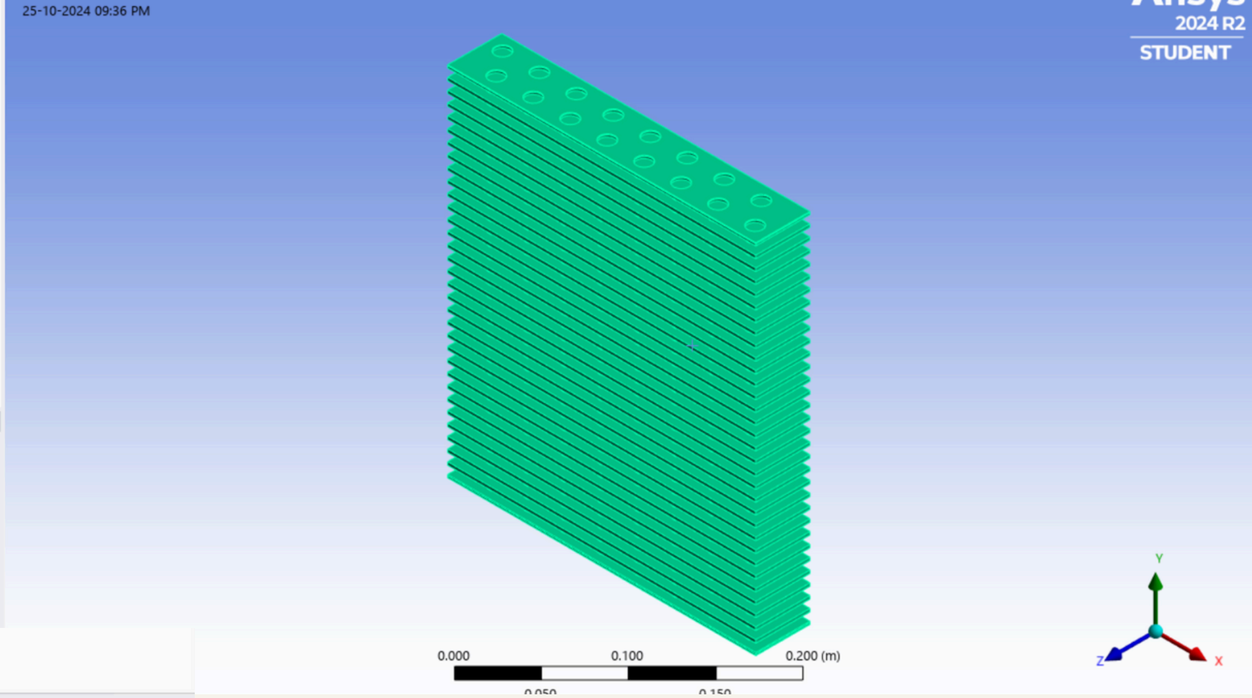
MESHING

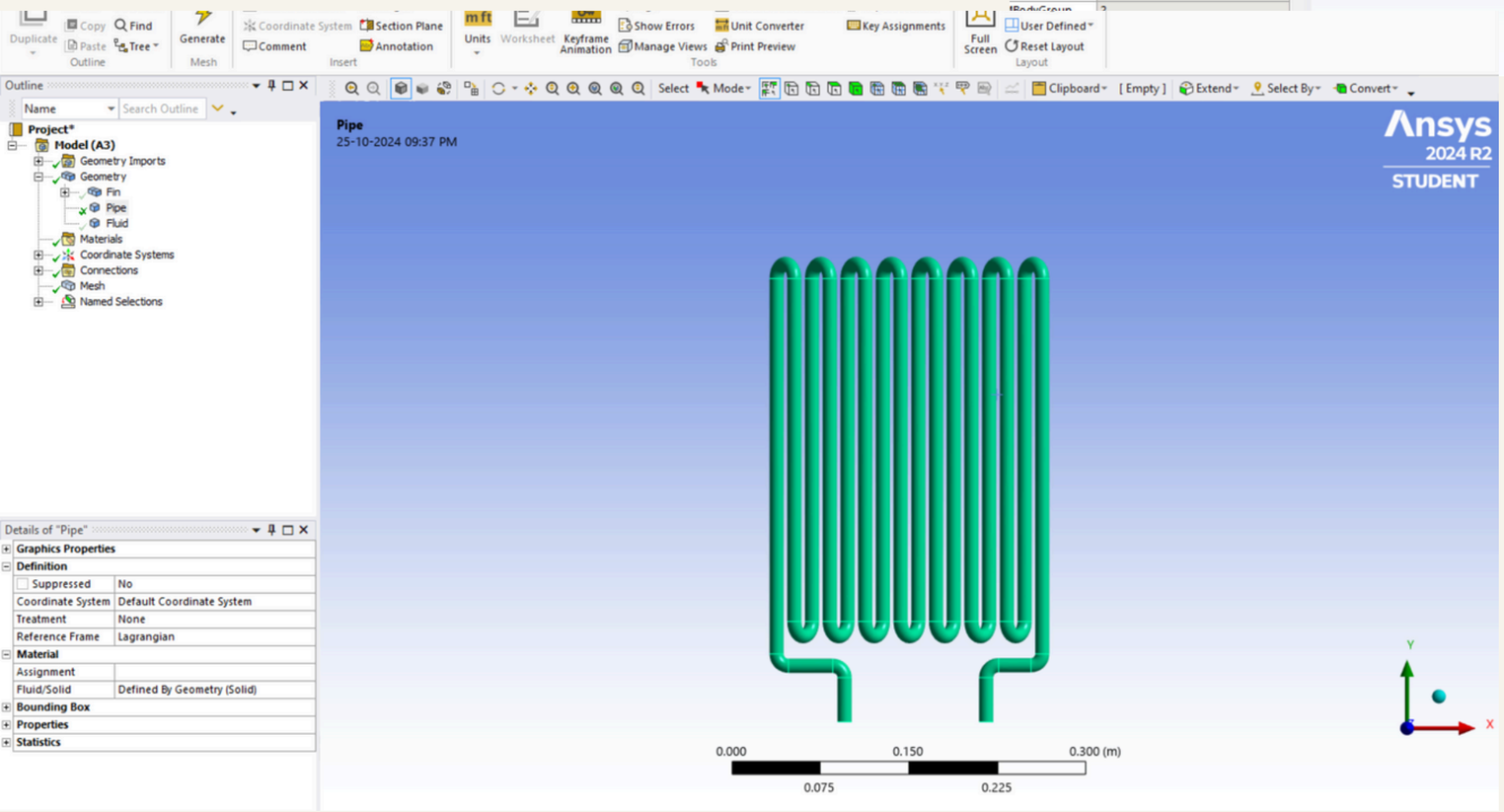

Outline
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Project*
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Geometry
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Named Selections
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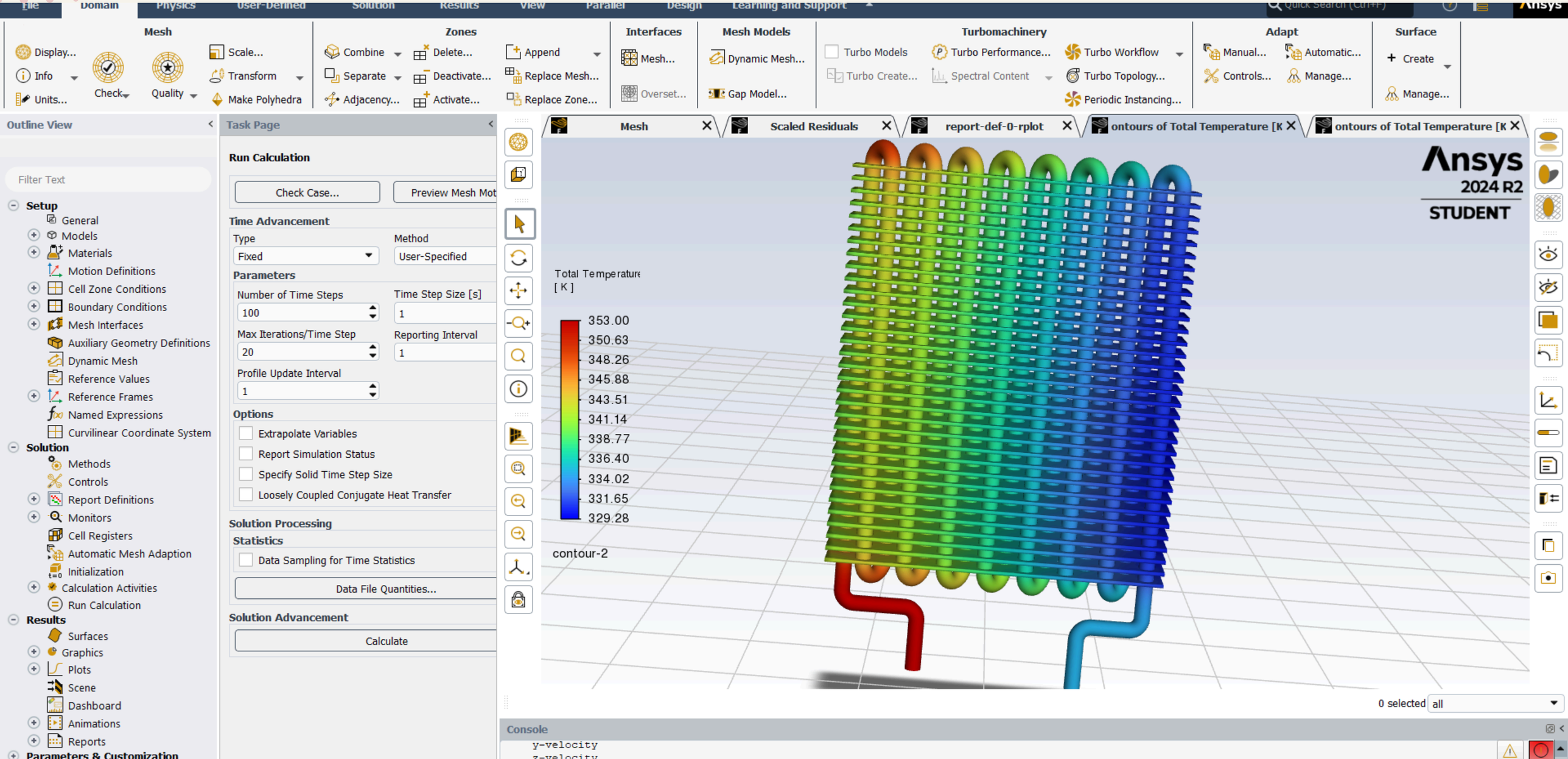
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Treatment None
Reference Frame Lagrangian
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Assignment
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Bounding Box
Properties
Statistics

PIPES

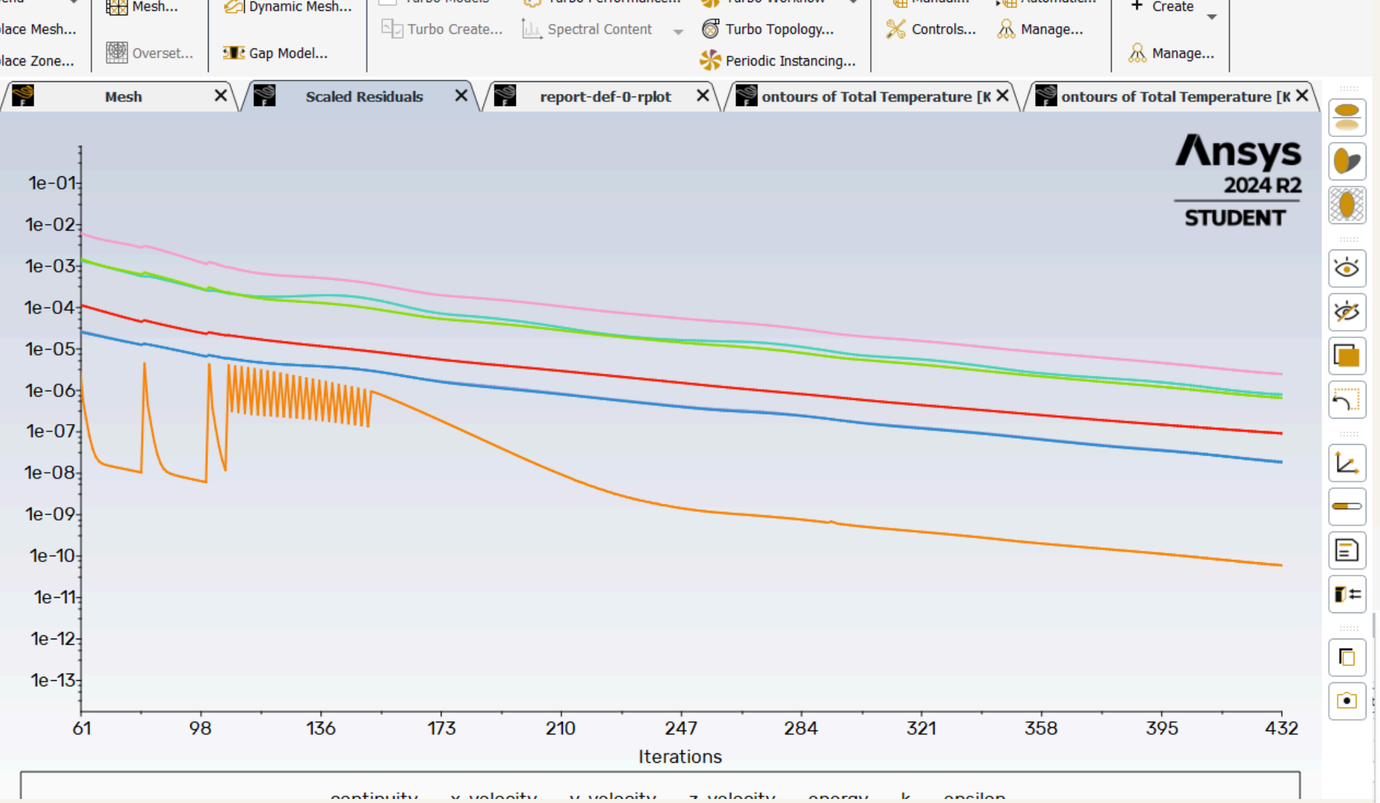
How radiators are used in aircraft and automobile industries?

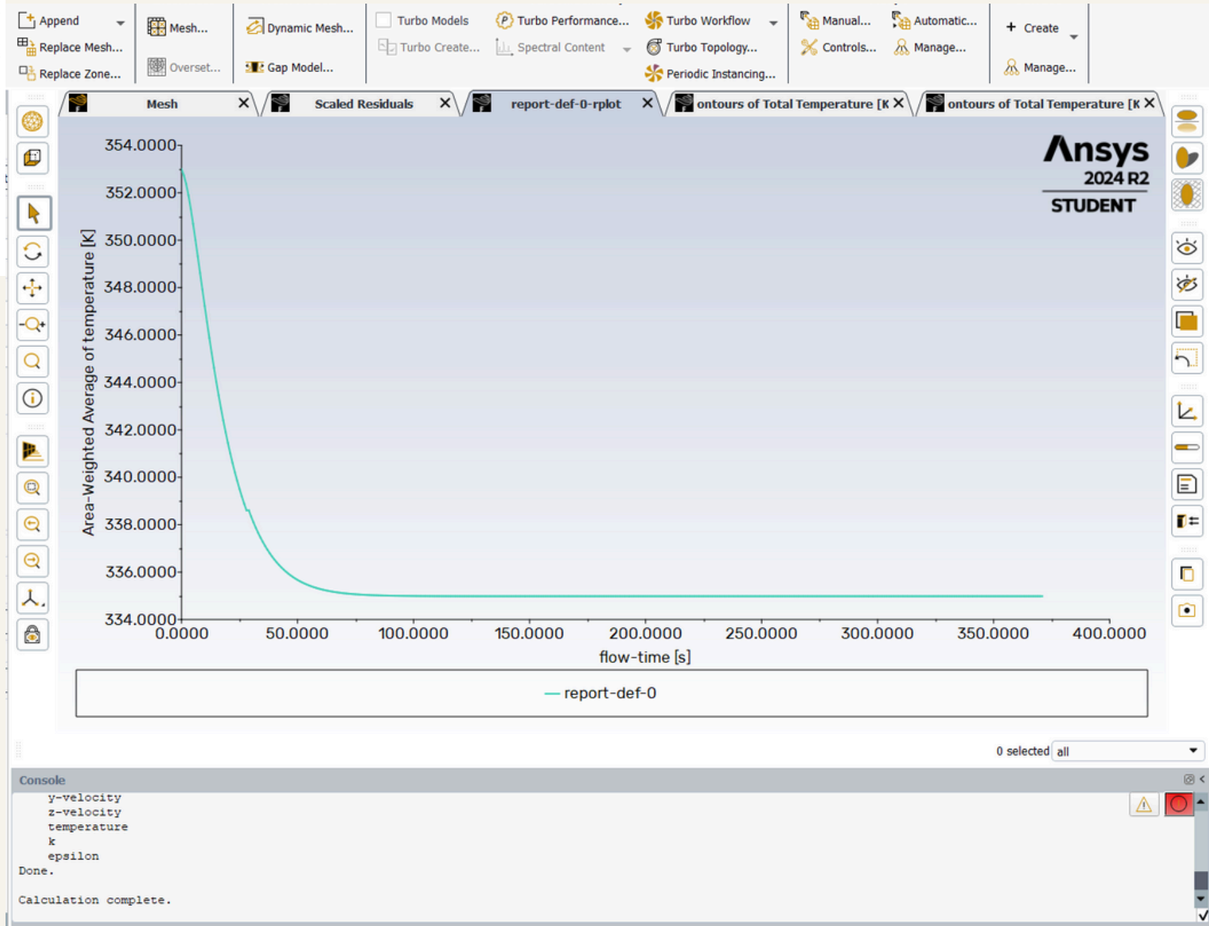
- Automobile Industry: In cars, radiators are a central component of the engine cooling system. They transfer the heat absorbed by the coolant away from the engine, allowing it to operate within an optimal temperature range. This prevents overheating, which can damage engine components and reduce fuel efficiency. Radiators in cars are typically air-cooled by the vehicle's movement, with fans providing additional airflow when needed. Advanced materials and design, such as lightweight aluminum and improved fin structures, enhance heat dissipation and support modern engine demands for greater fuel efficiency and reduced emissions.
- Aircraft Industry: Aircraft radiators, especially in piston-engine aircraft, serve a similar function but face additional challenges due to high-speed and high-altitude operation. They must withstand extreme temperatures and pressures while being compact and lightweight. Aircraft radiators are often liquid-cooled, using highly efficient fin structures and low-weight materials like aluminum to dissipate heat from engine components. Additionally, modern jet engines and turboprops use oil coolers and heat exchangers (similar to radiators) to manage the heat from engine oil, hydraulic fluids, and other components. Radiators in aircraft are carefully designed to minimize drag while maximizing heat transfer, critical for high-efficiency and fuel-saving designs in the aviation industry.

SOLUTION (WATER AND COPPER)

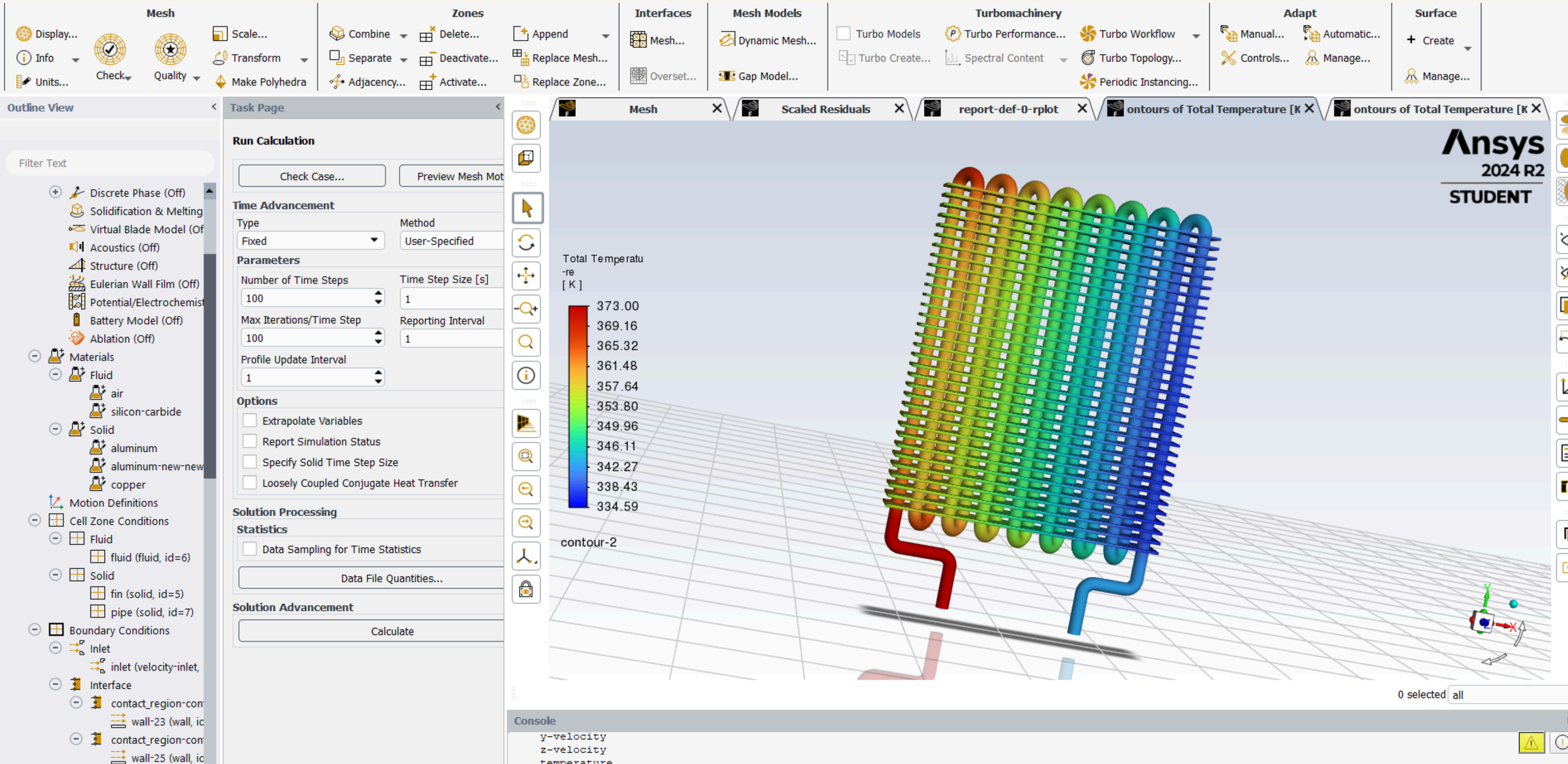


GRAPH

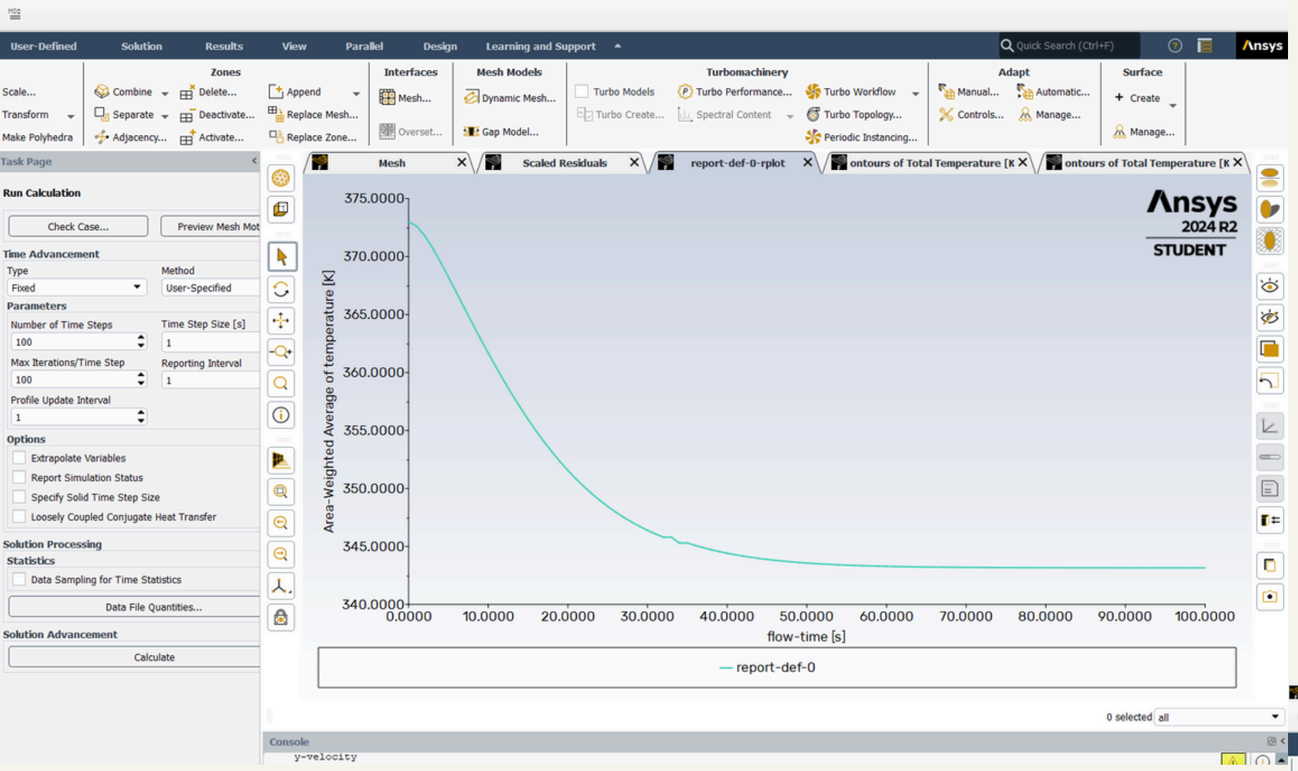

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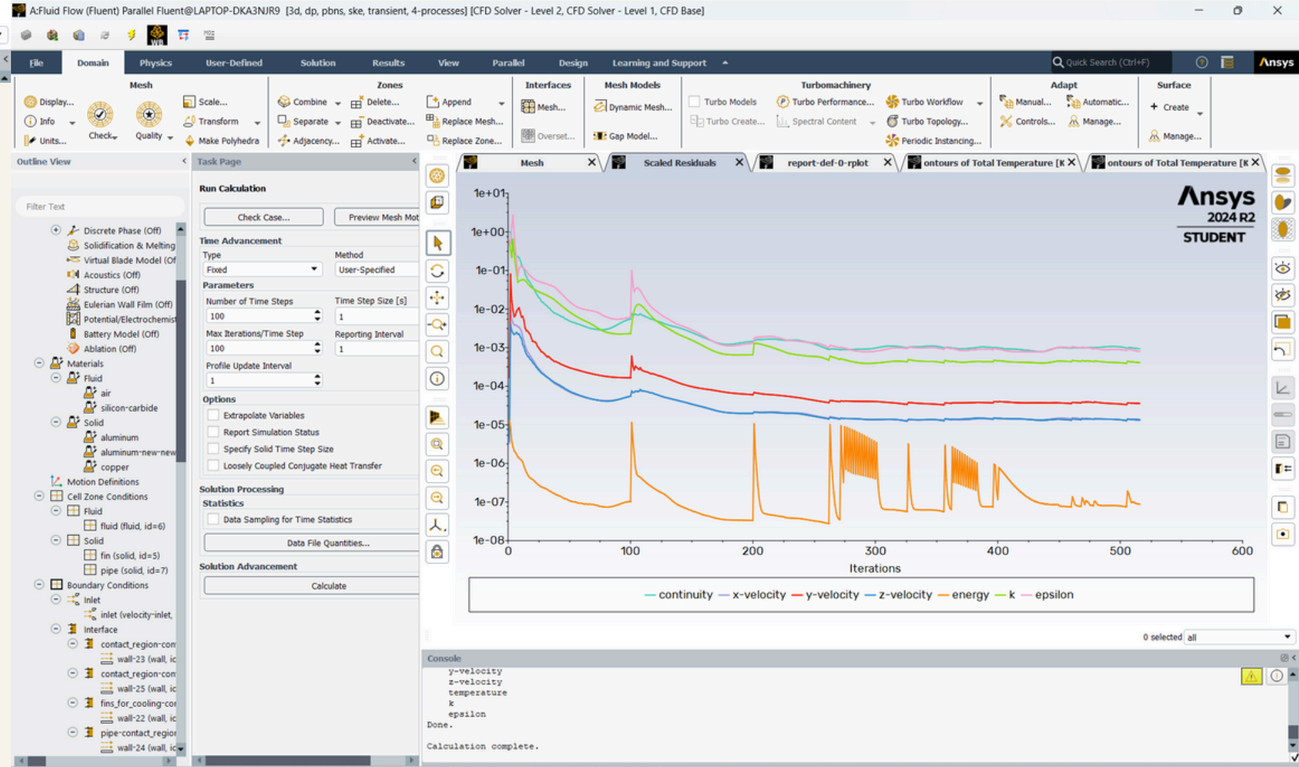

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Dynamic Mesh...
Gap Model...
Turbo Models
Turbo Create...
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SOUTION (SIC AND ALUMINIUM)



GRAPH


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A:Fluid Flow (Fluent) Parallel Fluent@LAPTOP-DKA3NJR9 [3d, dp, pbns, ske, transient, 4-processes] [CFD Solver - Level 2, CFD Solver - Level 1, CFD Base]
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Iterations
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x-velocity
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THANK YOU